



Data Engineering

SETUP POSTGRESQL & DBEAVER



pgAdmin

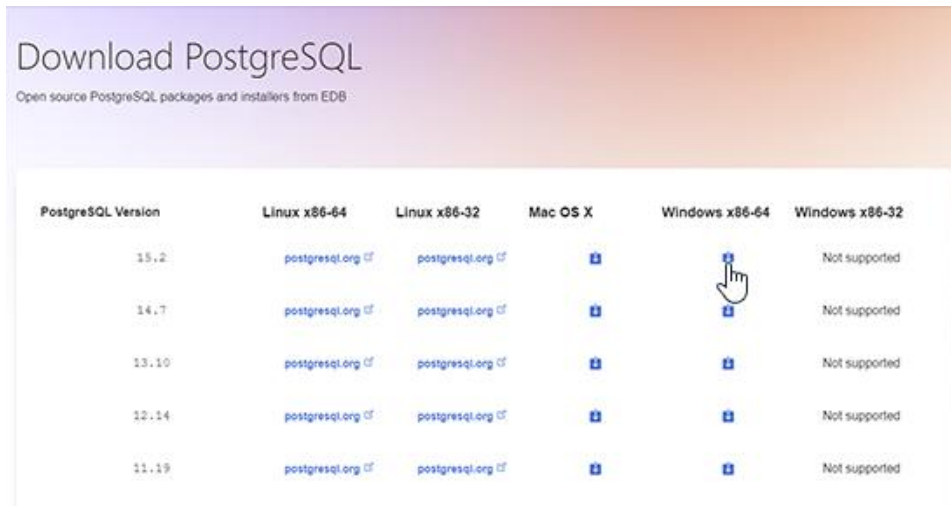


What is PostgreSQL?

🐘 PostgreSQL is a powerful database that remembers everything so you don't have to—unless you forget to back it up. 😊

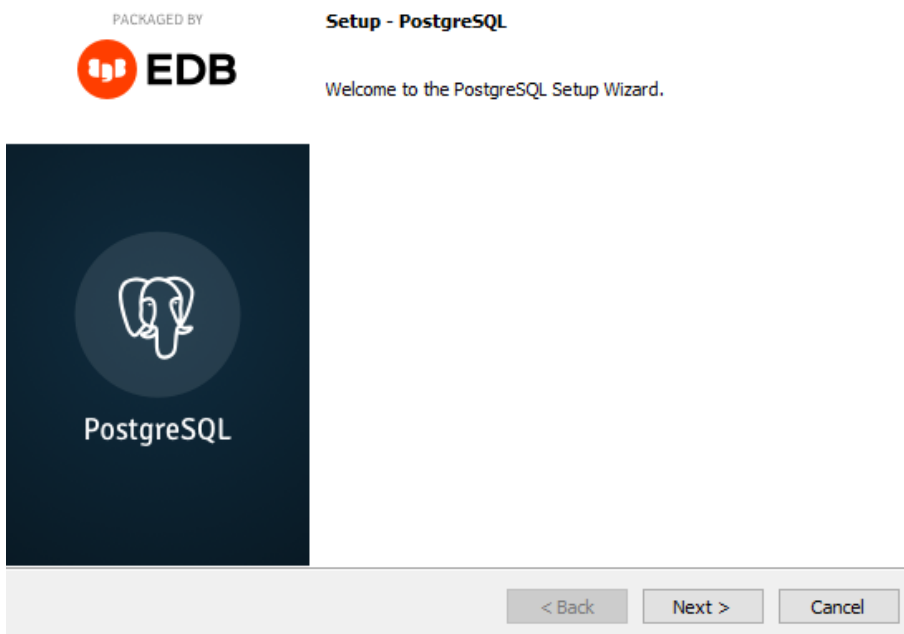
How to Install PostgreSQL on Windows

- To install PostgreSQL locally on your computer, visit the [installer by EDB](#), and download the newest version compatible with your operating system.
- I will choose the newest Windows version.



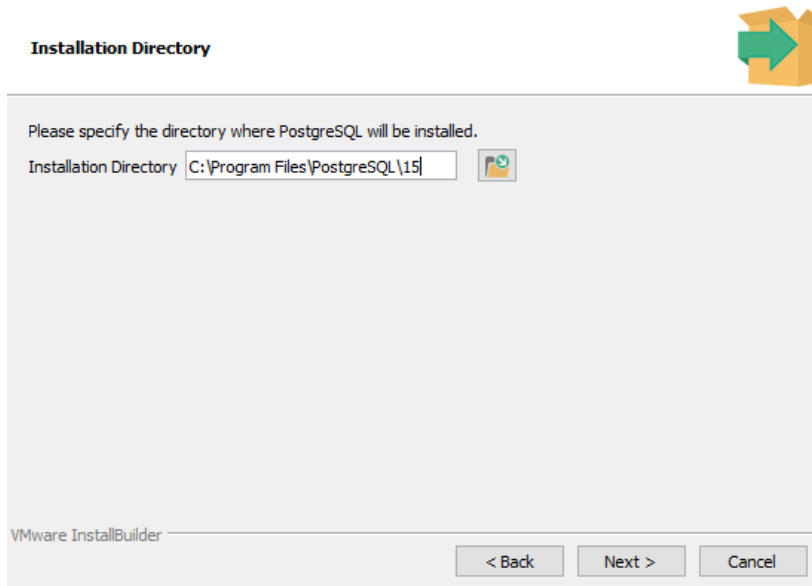
Start the Install

When the downloading is complete, double click the downloaded file and start the installation:



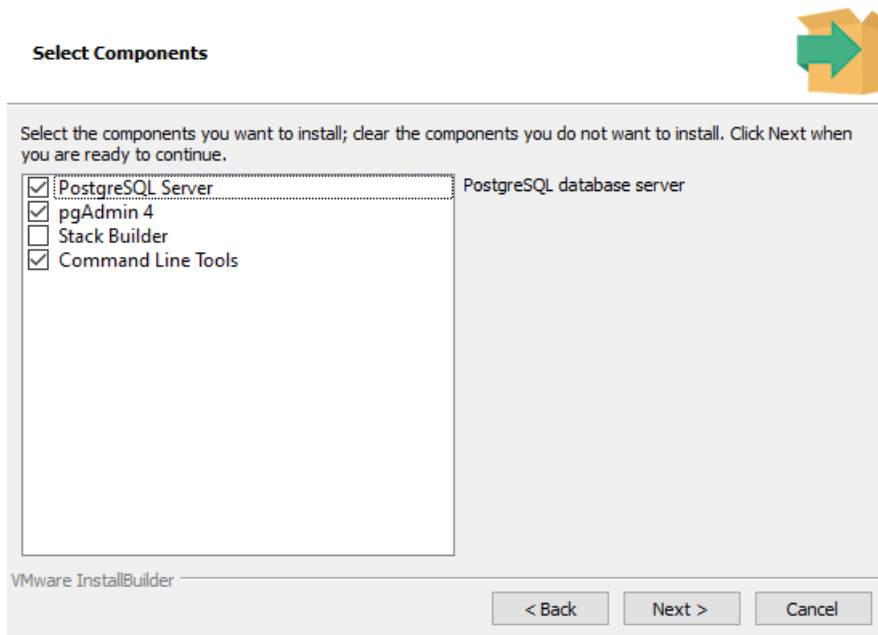
Specify Directory

You can specify the location of PostgreSQL, I will go with the default choice:



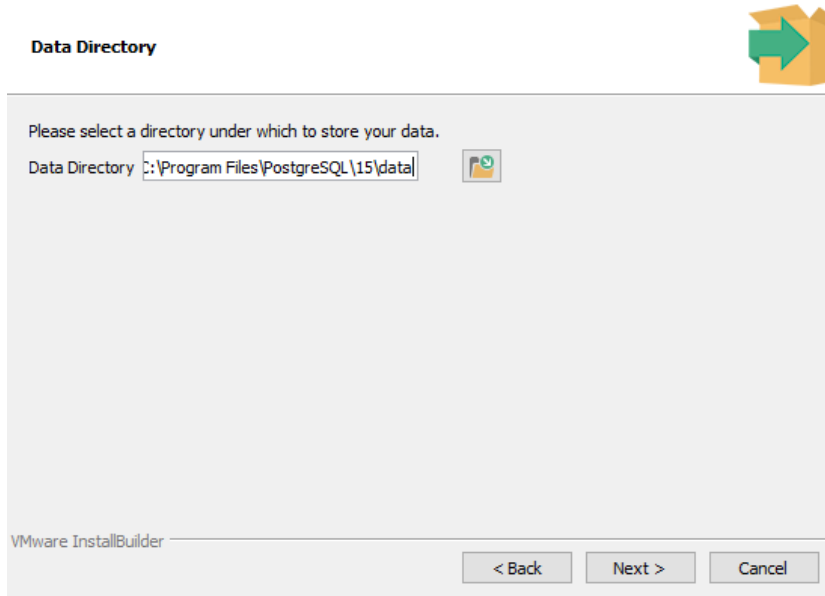
Select Components

To use PostgreSQL, you will need to install the PostgreSQL Server. In this tutorial we will also use the pgAdmin 4 component, and the Command Line Tools:



Storage Directory

You can also choose where to store the database data, I will go with the default choice:



Data Directory

Please select a directory under which to store your data.

Data Directory 

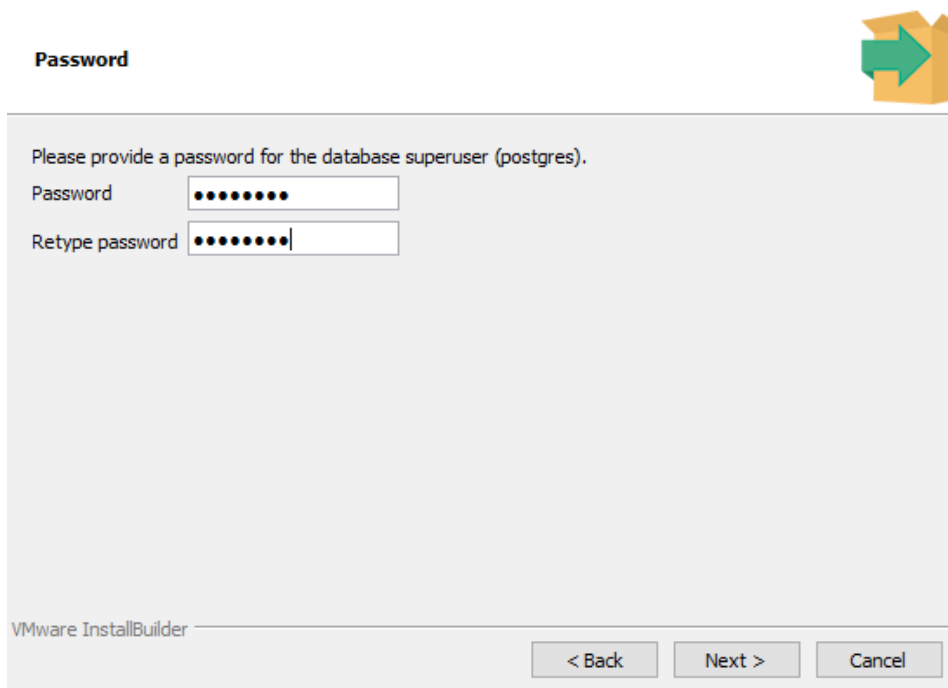
VMware InstallBuilder

< Back Next > Cancel

This screenshot shows a window titled 'Data Directory' from VMware InstallBuilder. It prompts the user to select a directory for storing database data. The 'Data Directory' field is pre-filled with 'C:\Program Files\PostgreSQL\15\data'. There is a small icon to the right of the text box, likely for browsing. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

Select Password

You will have to select a password to get access to the database. Since this is a local database, with no incoming connection, I will choose the password 12345678:



Password

Please provide a password for the database superuser (postgres).

Password

Retype password

VMware InstallBuilder

< Back Next > Cancel

This screenshot shows a window titled 'Password' from VMware InstallBuilder. It prompts the user to provide a password for the database superuser (postgres). There are two password input fields: 'Password' and 'Retype password'. Both fields are filled with masked characters (dots). At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

Select Port

You can set the port the server should listen on, I will go with the default choice:

Port 

Please select the port number the server should listen on.

Port

VMware InstallBuilder

< Back Next > Cancel

Select Locale

Select the geographically location of the database server:

Advanced Options 

Select the locale to be used by the new database cluster.

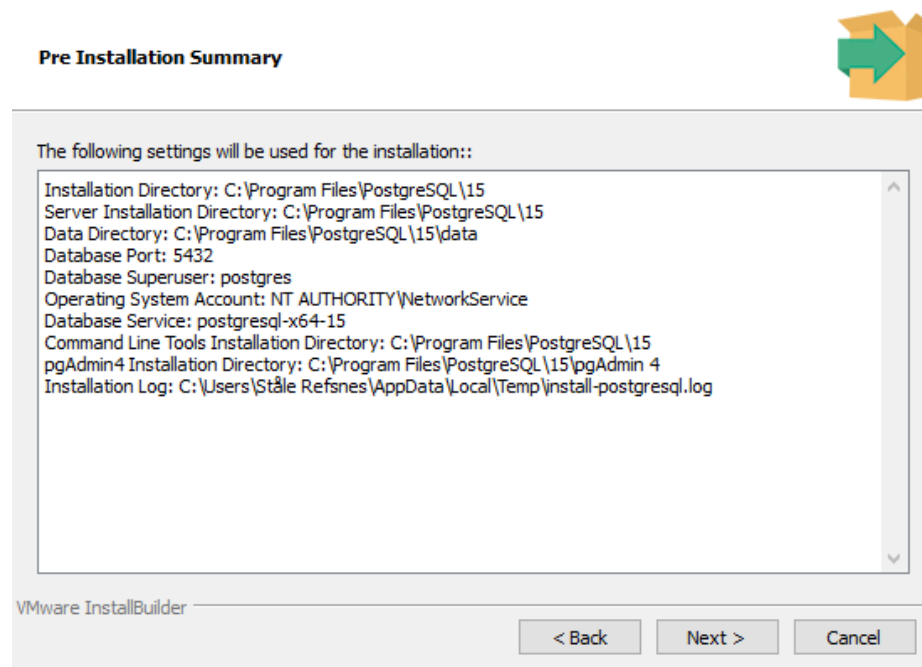
Locale

VMware InstallBuilder

< Back Next > Cancel

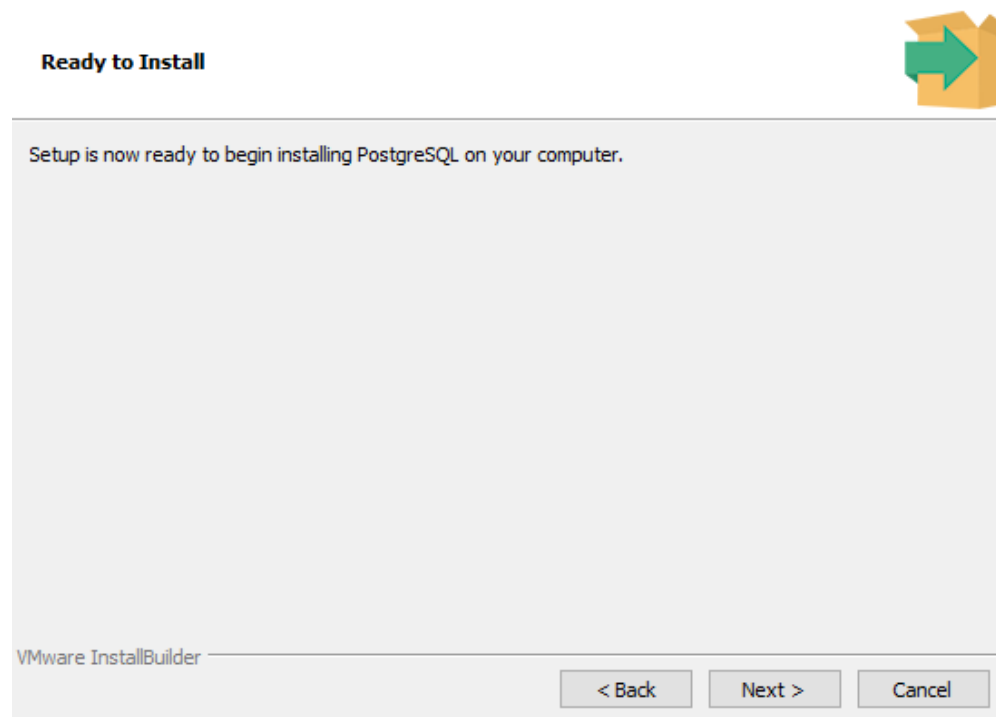
Final Check

If everything looks OK, click 'Next' to continue:



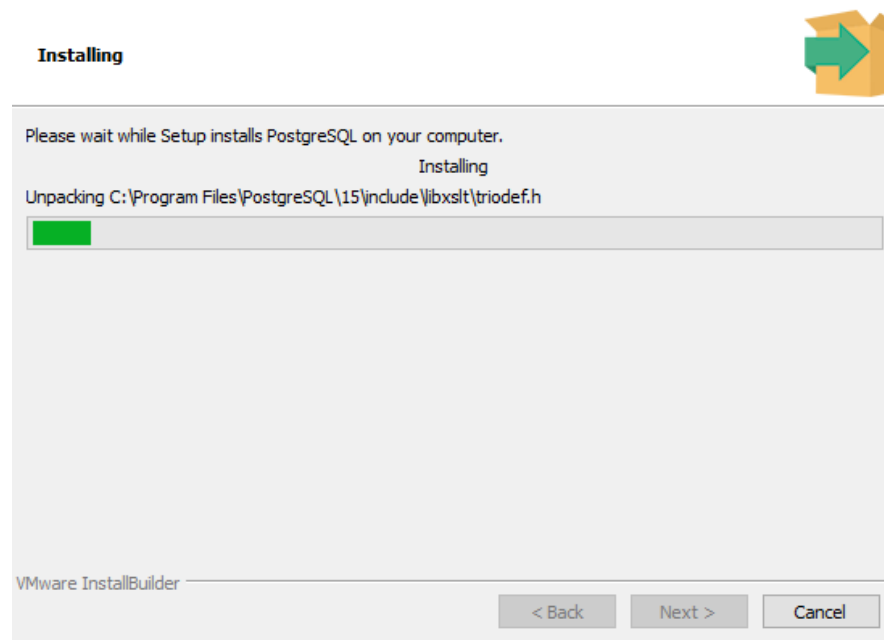
Start Installation:

Click 'Next' to start the installation:



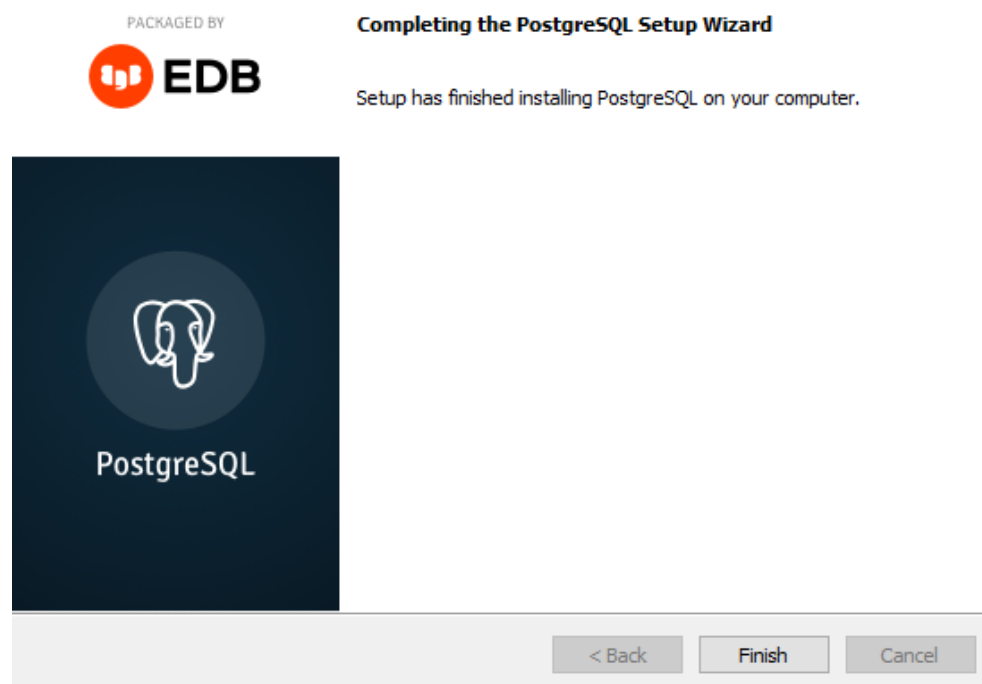
Installing

This can take a while, please wait.



Complete!

Now you have installed PostgreSQL on your computer, and in the next chapter you will start using it!



🍏 How to Install PostgreSQL on macOS

➤ Download PostgreSQL installer for macOS

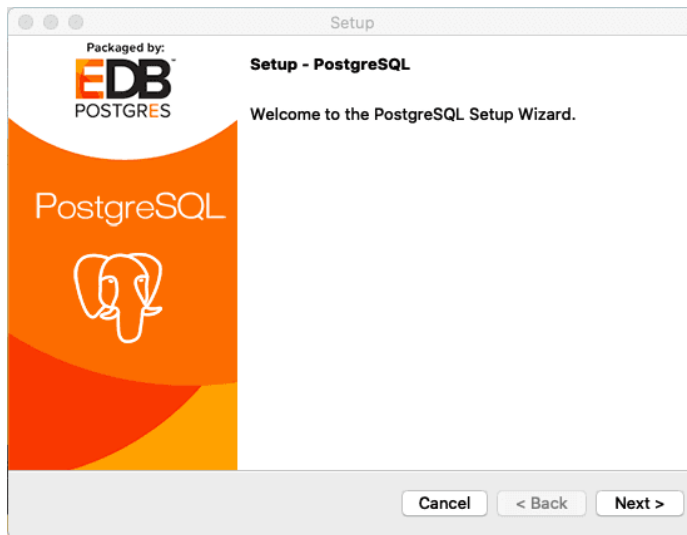
To download the PostgreSQL installer, you follow these steps:

- First, visit the [PostgreSQL installer download page](#).
- Then, download the PostgreSQL for macOS.

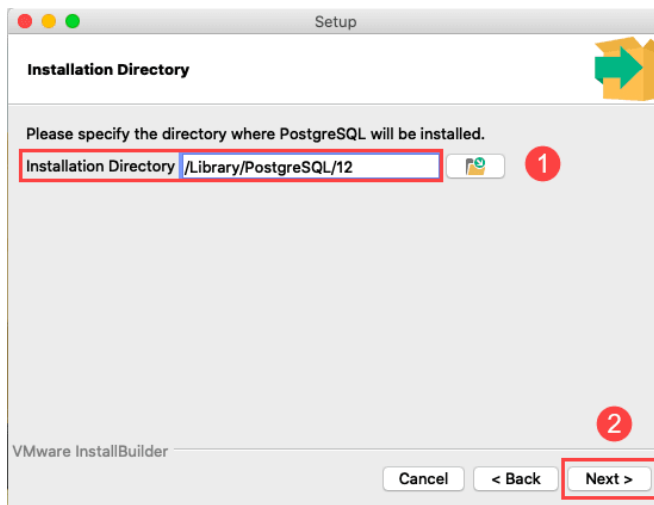
➤ Install PostgreSQL on macOS

To install PostgreSQL on macOS, you follow these steps:

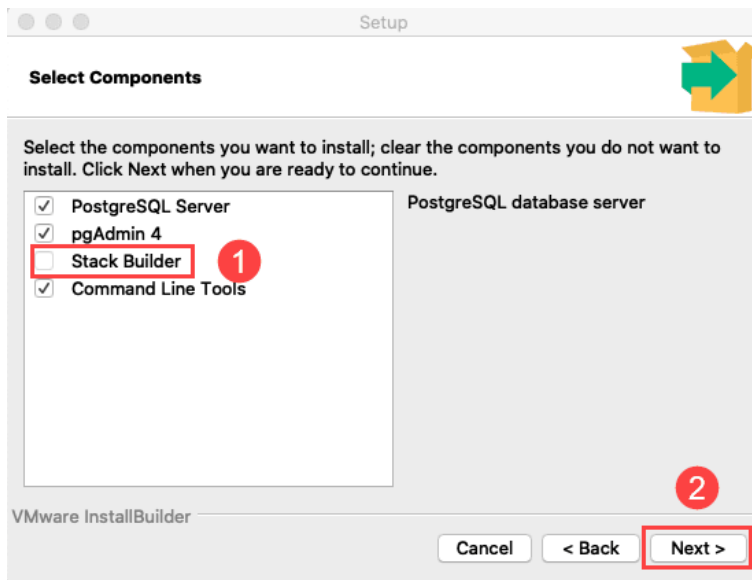
First, launch the setup wizard by double-click the installer file:



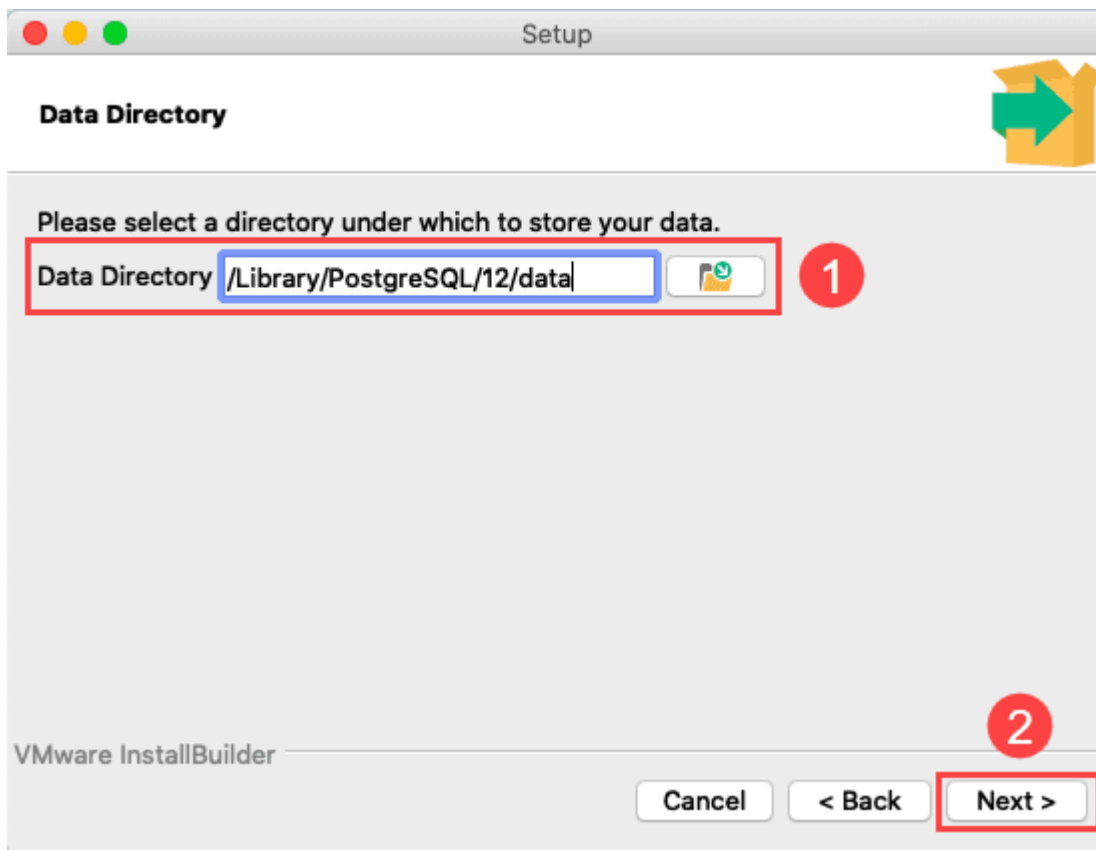
Second, select the directory where the PostgreSQL will be installed and click the Next button:



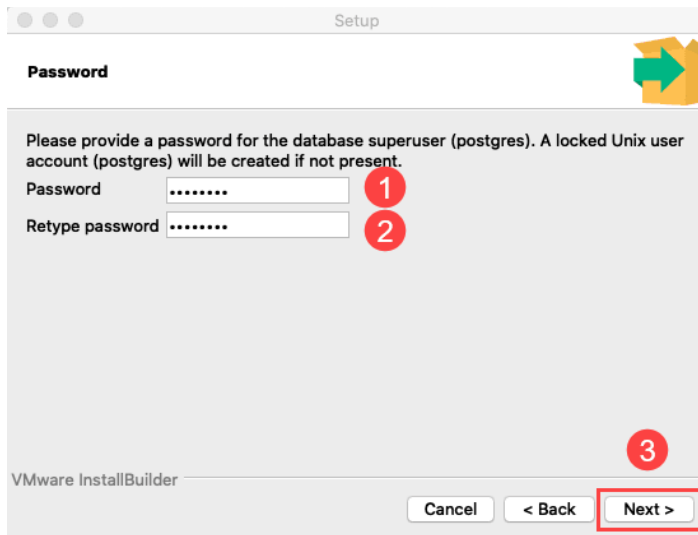
Third, select the components that you want to install, uncheck the Stack Builder, and click the Next button:



Fourth, specify a directory where PostgreSQL stores the data and click the Next button:

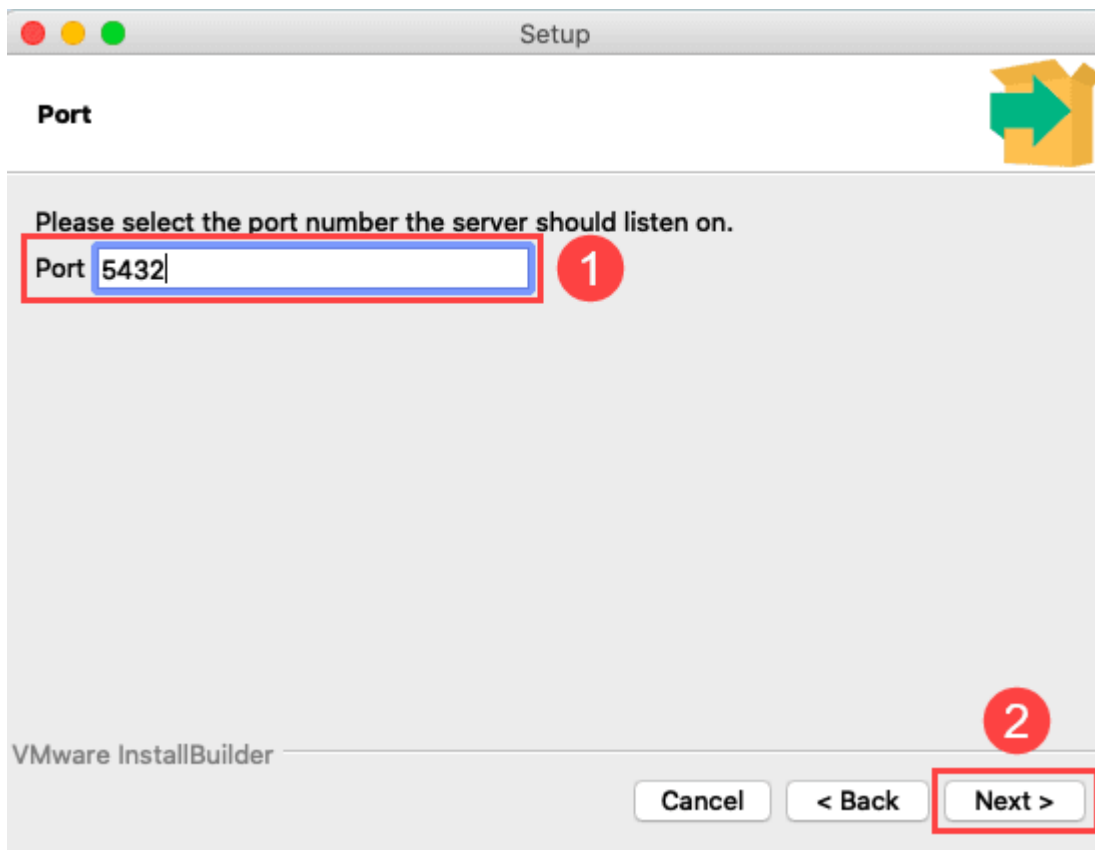


Fifth, enter the password for the postgres user account. You should note this password for logging in to the PostgreSQL database server later. After that, click the Next button.



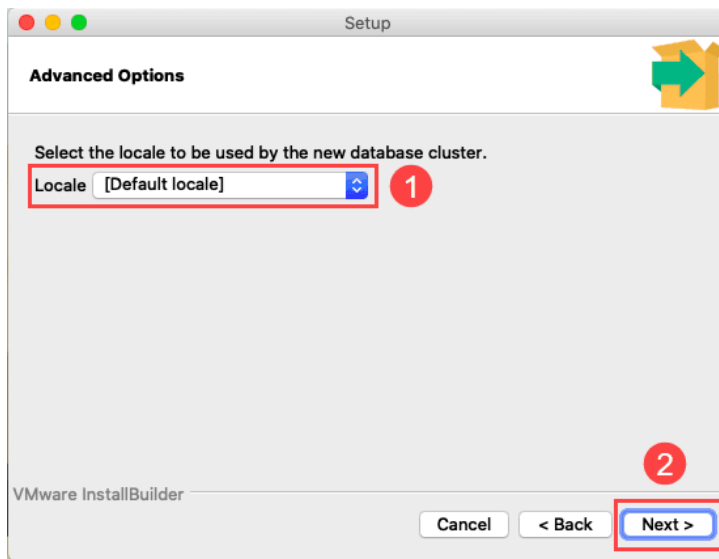
The screenshot shows a window titled "Setup" with a "Password" section. It contains two text input fields: "Password" and "Retype password", both filled with dots. Red circles with numbers 1 and 2 are placed next to these fields. A red circle with the number 3 is placed next to the "Next >" button at the bottom right. The text "Please provide a password for the database superuser (postgres). A locked Unix user account (postgres) will be created if not present." is displayed above the fields. The "VMware InstallBuilder" logo is in the bottom left, and "Cancel", "< Back", and "Next >" buttons are in the bottom right.

Sixth, specify the port number on which the PostgreSQL server will listen. By default, PostgreSQL uses port number 5432.

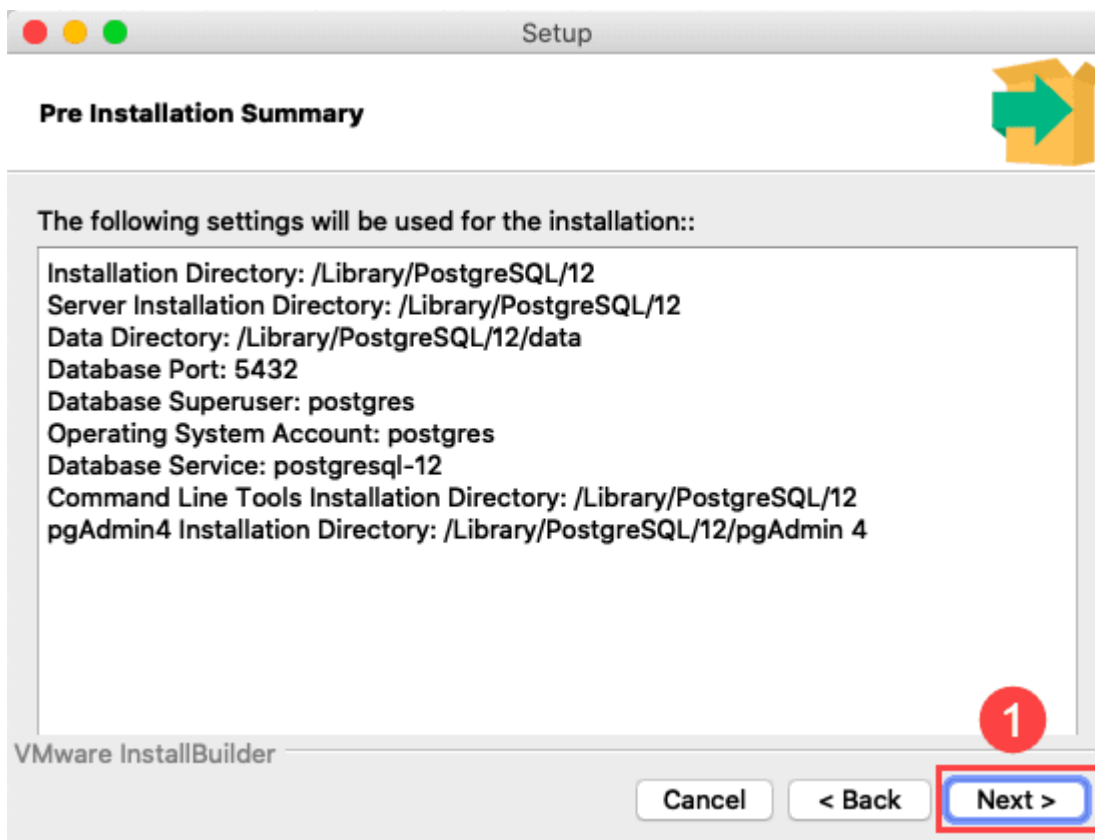


The screenshot shows a window titled "Setup" with a "Port" section. It contains a text input field labeled "Port" with the value "5432" entered. A red circle with the number 1 is placed next to the input field. A red circle with the number 2 is placed next to the "Next >" button at the bottom right. The text "Please select the port number the server should listen on." is displayed above the field. The "VMware InstallBuilder" logo is in the bottom left, and "Cancel", "< Back", and "Next >" buttons are in the bottom right.

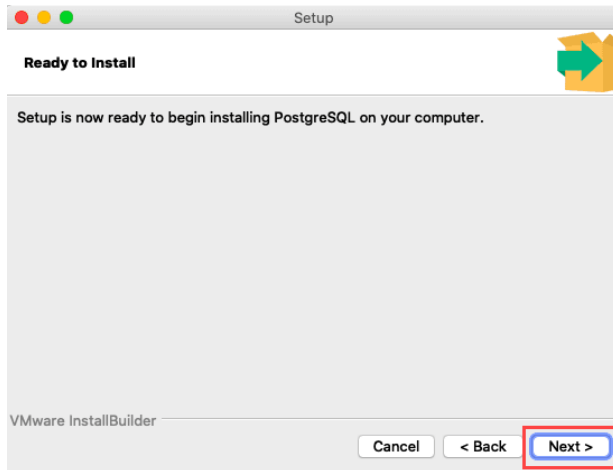
Seventh, select the locale used by PostgreSQL. By default, PostgreSQL uses the locale of the current operating system:



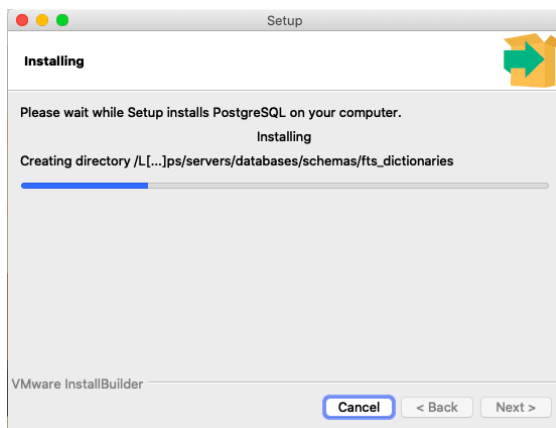
Eighth, review the installation information. If everything looks correct, click the Next button to begin the installation.



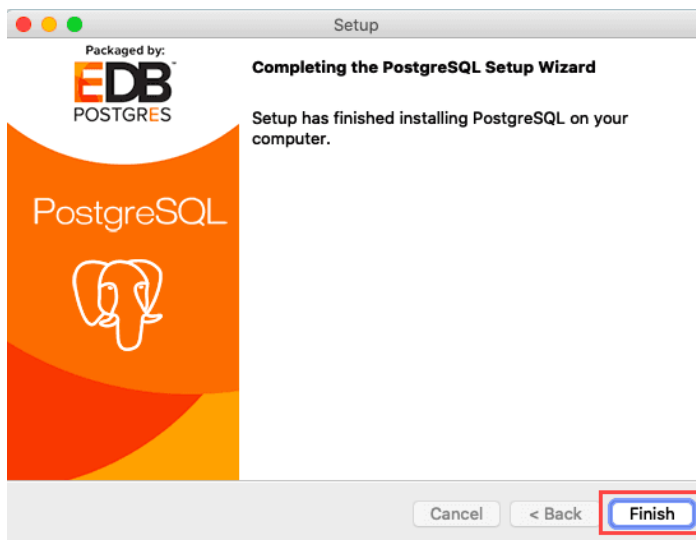
Ninth, click the Next button to start installing the PostgreSQL database server on your computer:



It will take a few minutes to complete the installation.



Finally, click the Finish button once the installation is completed:



Installing DBeaver

Downloading DBeaver

1. **Visit the DBeaver Website:** Head over to the official [DBeaver download page](#).
2. **Choose the Right Edition:** DBeaver offers two main editions: Community Edition (free) and Enterprise Edition (paid with additional features). Select the edition that best fits your needs.
3. **Download the Installer:** Click the download link for your operating system to save the installer file.

Installing on Windows

1. **Run the Installer:** Locate the downloaded .exe file and double-click it to start the installation process.

DBeaver-x.y.z-Windows-x86_64.exe

2. **Follow the Installation Wizard:** The installation wizard will guide you through several steps. Accept the license agreement, choose the installation directory, and select additional components if needed.
3. **Complete the Installation:** Click "Finish" when the installer informs you that the installation has completed.

Installing on Mac

1. **Open the DMG File:** Double-click the downloaded .dmg file.

DBeaver-x.y.z-macos.dmg

2. **Drag DBeaver to Applications:** In the opened window, drag the DBeaver icon to your Applications folder.
3. **Launch the Application:** Open DBeaver from the Applications folder. You may need to authorize the app in your security settings on first launch.